BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

PHYSICS — SSC-I (9th Grade)

Syllabus: Unit 1 — Physical Quantities & Measurement | Unit 2 — Kinematics | Unit 3 — Dynamics-I | Unit 4 — Dynamics-II | Unit 5 — Pressure & Deformation in Solids | Unit 6 — Work & Energy (till 6.4) | Unit 9 — Nature of Science and Physics

Total Marks: 65 | **Time Allowed: 2 hours 30 minutes** | **Version: 1**

General Instructions:

1. This paper has **three sections**: A (MCQs), B (Short Questions), C (Long Questions).
2. **Section A** (12 marks): 12 MCQs. Mark the correct bubble. Time: 15 minutes.
3. **Section B** (33 marks): 11 short questions × 3 marks each. Answer *either* the main question *or* the OR alternative.
4. **Section C** (20 marks): 4 long questions × 5 marks each. Answer *either* the main question *or* the OR alternative. Use g = 10 m/s² in numerical problems unless otherwise specified.

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks) [15 min]

Q.1: Choose the correct answer and fill in the corresponding bubble.

#	Question	A	B	C	D	Answer
1	Which of the following is a base physical quantity?	Force	Electric current	Density	Velocity	ⒶⒷⒸⒹ
2	The pascal (Pa), SI unit of pressure, is dimensionally equivalent to:	kg/m^3	$kg \cdot m/s^2$	$kg/(m \cdot s^2)$	$kg \cdot m^2/s^2$	ⒶⒷⒸⒹ
3	The gradient of a distance-time graph gives the object's:	Acceleration	Displacement	Instantaneous speed	Average acceleration	ⒶⒷⒸⒹ
4	A ball dropped from rest falls freely for 7 s. Taking $g = 10\ m/s^2$, its velocity at impact is:	17 m/s	35 m/s	700 m/s	70 m/s	ⒶⒷⒸⒹ
5	Which fundamental force is strongest, though it acts only over nuclear-sized distances?	Gravity	Weak force	Electromagnetic	Strong force	ⒶⒷⒸⒹ
6	A jumper's feet push down on the ground (action); the ground pushes back (reaction). This illustrates:	Newton's 1st law	Newton's 2nd law	Newton's 3rd law	Conservation of mass	ⒶⒷⒸⒹ
7	A force of 35 N acts perpendicular to a spanner, 0.5 m from a bolt. The torque produced is:	17.5 N·m	70 N·m	0.014 N·m	35.5 N·m	ⒶⒷⒸⒹ
8	A pencil balanced on its tip topples over completely when slightly disturbed. This illustrates:	Stable	Unstable	Neutral	Dynamic	ⒶⒷⒸⒹ
9	Lines on a weather map joining places of equal atmospheric pressure are called:	Isotherms	Isobars	Contours	Isotopes	ⒶⒷⒸⒹ
10	Pressure at the bottom of a 7 m deep water column ($\rho = 1000\ kg/m^3$, $g = 10\ m/s^2$) is:	7,000 Pa	700 Pa	70,000 Pa	700,000 Pa	ⒶⒷⒸⒹ
11	A force of 60 N at 60° to the horizontal moves a crate 5 m. Taking cos 60° = 0.5, work done is:	300 J	30 J	150 J	600 J	ⒶⒷⒸⒹ
12	Which branch of physics studies the origin, development and future of the entire universe?	Astronomy	Cosmology	Astrophysics	Thermal Physics	ⒶⒷⒸⒹ

SECTION B — Short Questions (11 × 3 = 33 Marks)

Q.2: Attempt **all** questions. Answer *either* the main question *or* the OR alternative.

Part	Question	Marks	OR	OR Question	Marks
(i)	Name the seven **SI base physical quantities**. State the SI unit for each of: (a) area, (b) velocity, (c) force.	**3**	**OR**	Differentiate between **physical quantities** and **non-physical quantities**. Give two examples of each.	**3**
	Differentiate between **speed** and **velocity**. A car moves at a constant speed of 40 m/s			Sketch distance-time graphs for (a) a body at rest, (b) uniformly increasing speed. What does	

(ii)	around a circular track. Is its velocity also constant? Give a reason.	**3**	**OR**	the gradient represent?	**3**
(iii)	A delivery van's speed increases uniformly from 8 m/s to 32 m/s in 6 s. Calculate (a) its acceleration, (b) the distance covered.	**3**	**OR**	A drone drops a package from a height of 320 m. Taking $g = 10\ m/s^2$, find (a) the time to reach the ground, (b) its impact velocity.	**3**
(iv)	Name the **four fundamental forces** of nature in decreasing order of strength. Which one holds the nucleus of an atom together?	**3**	**OR**	State **Newton's second law** of motion. A resultant force of 45 N produces an acceleration of $5\ m/s^2$ in a body. Calculate the mass of the body.	**3**
(v)	Differentiate between a **system diagram** and a **free-body diagram (FBD)**. Draw an FBD for a crate pulled at constant velocity by a rope across a rough floor.	**3**	**OR**	Define **net (resultant) force**. Two people push a stalled car in the same direction with forces 18 N and 25 N. Find the net force and name the type of parallel forces.	**3**
(vi)	Identify the type of **equilibrium** (stable, unstable, neutral) shown by: (a) a ball in a bowl, (b) a pencil balanced on its tip, (c) a ball rolled to rest on a flat surface. Give a reason for each.	**3**	**OR**	Explain **terminal velocity**. Describe how a skydiver's weight and air resistance change until she reaches it.	**3**
(vii)	State two **advantages** and two **disadvantages** of friction. Name one method used to reduce friction between machine parts.	**3**	**OR**	Define **centripetal force** and write its formula. A 0.6 kg ball is whirled in a horizontal circle of radius 0.9 m at 6 m/s. Calculate the centripetal force.	**3**
(viii)	Explain how **atmospheric pressure** relates to weather. What are **isobars**, and what do closely-packed isobars indicate?	**3**	**OR**	A tank has a 2 m layer of petrol ($700\ kg/m^3$) on a 3 m layer of water ($1000\ kg/m^3$). Taking $g = 10\ m/s^2$, find (a) the pressure due to the petrol layer, (b) the total pressure at the bottom.	**3**
(ix)	Define **pressure** and its SI unit. A snowmobile's runners exert 4500 N over a contact area of $0.25\ m^2$. Calculate the pressure exerted.	**3**	**OR**	State **Pascal's principle**. Using a balloon being inflated as an example, explain how pressure is transmitted equally in an enclosed fluid.	**3**
(x)	Define **work done** ($W = Fd$). A child pulls a toy wagon with a rope at 40° above horizontal, moving it 6 m. Taking $\cos 40° = 0.77$, calculate the work done.	**3**	**OR**	Describe how **fossil fuels** form over millions of years. Name the three main types and state one environmental disadvantage of burning them.	**3**
(xi)	Differentiate **Physical Sciences** from **Biological Sciences**, giving two branches of each. Which category does Physics belong to?	**3**	**OR**	Physics is an **interdisciplinary field**. Give two examples (other than Biophysics) of how physics principles apply to other disciplines.	**3**

SECTION C — Long Questions (4 × 5 = 20 Marks)

Q.3: Attempt **all** questions. Answer *either* the main question *or* the OR alternative.

Q.	Question	Marks	OR	OR Question	Marks
3(i)	(a) Derive $v = u + at$ for uniformly accelerated motion. [2] (b) A van's speed increases uniformly from 7 m/s to 27 m/s over 136 m. Find (i) acceleration, (ii) time taken. [3]	**5**	**OR**	(a) State the three equations of motion, defining each symbol. [2] (b) A coconut falls from a tree and takes 2.5 s to hit the ground. Taking $g = 10\ m/s^2$, find (i) its impact velocity, (ii) the height of the tree. [3]	**5**
3(ii)	(a) State Newton's three laws, with one example each (not a vehicle or firearm). [2] (b) Draw an FBD for a lift accelerating upward, labelling the forces. [1] (c) A lift of mass 800 kg accelerates upward at $2\ m/s^2$. Taking $g = 10\ m/s^2$, find the cable tension. [2]	**5**	**OR**	(a) State the law of conservation of momentum. [1] (b) A 1000 kg car at 18 m/s collides head-on with a 1500 kg truck at 6 m/s (opposite direction) and they lock together. Taking the car's direction as positive, find their common velocity. [2] (c) Find the loss in kinetic energy during the collision. [2]	**5**
3(iii)	(a) Show that $P = \rho gh$ for a liquid column, using $P = F/A$. [2] (b) A dam wall holds back water to a depth of 18 m ($\rho = 1000\ kg/m^3$, $g = 10\ m/s^2$). Find the water pressure at the base. [2] (c) The dam base has an area of $40\ m^2$ in contact with water. Find the total force on it. [1]	**5**	**OR**	(a) State Hooke's law and define spring constant, with its SI unit. [2] (b) A spring extends 5 cm under a 10 N load, obeying Hooke's law up to a 12 cm extension. Find (i) the spring constant, (ii) the maximum load within this limit. [3]	**5**
3(iv)	(a) State the law of conservation of energy. [1] (b) A 6 kg rock is dropped from a 45 m cliff ($g = 10\ m/s^2$, no air resistance). Find (i) its PE at the top, (ii) its KE and speed at impact. [3] (c) If air resistance limits its actual impact speed to 24 m/s, find the work done against air resistance. [1]	**5**	**OR**	(a) Describe how fossil fuels form, and name the three main types. [2] (b) Explain nuclear fission, naming the two materials used as nuclear fuel in power plants. [2] (c) State one advantage and one disadvantage of nuclear fuel. [1]	**5**

Invigilator's Signature: ________________ Candidate's Signature: ________________

Chapters Covered: Units 1–6 (till 6.4) & 9 **— End of Question Paper —**